

Training Verifiers to Solve Math Word Problems

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Abstract

State-of-the-art language models can match human performance on many tasks, but they still struggle to robustly perform multi-step mathematical reasoning. To diagnose the failures of current models and support research, we introduce GSM8K, a dataset of 8.5K high quality linguistically diverse grade school math word problems. We find that even the largest transformer models fail to achieve high test performance, despite the conceptual simplicity of this problem distribution. To increase performance, we propose training verifiers to judge the correctness of model completions. At test time, we generate many candidate solutions and select the one ranked highest by the verifier. We demonstrate that verification significantly improves performance on GSM8K, and we provide strong empirical evidence that verification scales more effectively with increased data than a finetuning baseline.

1 Introduction

In recent years, large language models have demonstrated impressive skills across many diverse tasks (Wang et al., 2019; Brown et al., 2020). Kaplan et al. (2020) describe the consistent benefits of increasing model size, characterizing scaling trends that hold across many orders of magnitude. However, even the largest models falter when required to perform multi-step mathematical reasoning (Hendrycks et al., 2021). Model samples frequently contain catastrophic mistakes, even after the model has been appropriately finetuned. Mathematical reasoning thus reveals a critical weakness in modern language models.

One significant challenge in mathematical reasoning is the high sensitivity to individual mistakes (Shen et al., 2021a). When generating a solution, autoregressive models have no mechanism to correct their own errors. Solutions that veer off-course quickly become unrecoverable. If we rely purely on generative methods and extrapolate from current trends, we will require an exorbitant

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Problem: Beth bakes 4, 2 dozen batches of cookies in a week. If these cookies are shared amongst 16 people equally, how many cookies does each person consume? Solution: Beth bakes 4 2 dozen batches of cookies for a total of $4 \times 2 = <<4 \times 2 = 8>>$ 8 dozen cookies There are 12 cookies in a dozen and she makes 8 dozen cookies for a total of $12 \times 8 = <<12 \times 8 = 96>>$ 96 cookies She splits the 96 cookies equally amongst 16 people so they each eat $96 / 16 = <<96 / 16 = 6>>$ 6 cookies Final Answer: 6
Problem: Mrs. Lim milks her cows twice a day. Yesterday morning, she got 68 gallons of milk and in the evening, she got 82 gallons. This morning, she got 18 gallons fewer than she had yesterday morning. After selling some gallons of milk in the afternoon, Mrs. Lim has only 24 gallons left. How much was her revenue for the milk if each gallon costs \$3.50? Mrs. Lim got 68 gallons - 18 gallons = $<<68 - 18 = 50>>$ 50 gallons this morning. So she was able to get a total of 68 gallons + 82 gallons + 50 gallons = $<<68 + 82 + 50 = 200>>$ 200 gallons. She was able to sell 200 gallons - 24 gallons = $<<200 - 24 = 176>>$ 176 gallons. Thus, her total revenue for the milk is $\\$3.50/\text{gallon} \times 176 \text{ gallons} = \\$<<3.50 \times 176 = 616>>$ 616. Final Answer: 616
Problem: Tina buys 3 12-packs of soda for a party. Including Tina, 6 people are at the party. Half of the people at the party have 3 sodas each, 2 of the people have 4, and 1 person has 5. How many sodas are left over when the party is over? Solution: Tina buys 3 12-packs of soda, for $3 \times 12 = <<3 \times 12 = 36>>$ 36 sodas 6 people attend the party, so half of them is $6 / 2 = <<6 / 2 = 3>>$ 3 people Each of those people drinks 3 sodas, so they drink $3 \times 3 = <<3 \times 3 = 9>>$ 9 sodas Two people drink 4 sodas, which means they drink $2 \times 4 = <<2 \times 4 = 8>>$ 8 sodas With one person drinking 5, that brings the total drank to $5 + 9 + 8 = <<5 + 9 + 8 = 25>>$ 25 sodas As Tina started off with 36 sodas, that means there are $36 - 25 = <<36 - 25 = 11>>$ 11 sodas left Final Answer: 11

Figure 1: Three example problems from GSM8K. Calculation annotations are highlighted in red.

parameter count to achieve even moderate performance on distributions as challenging as the MATH dataset (Hendrycks et al., 2021). This evidence strongly motivates the search for methods with more favorable scaling laws.

We propose training verifiers to evaluate the correctness of model generated solutions, similar to concurrent work by Shen et al. (2021a). At test time, we sample a fixed number of candidate solutions and select the solution ranked highest by the verifier. Verifiers benefit both from their inherent optionality and from verification being a simpler task than generation in general.

To facilitate research, we are releasing GSM8K, a dataset of 8.5K high quality problems at the grade school math level. We designed this dataset to have high linguistic diversity while relying on relatively simple grade school math concepts. State-of-the-art language models struggle to achieve high performance on this dataset, primarily due to the high diversity among problems. At the same time, GSM8K solutions depend only on elementary concepts, so achieving high test performance is a tractable goal.

Our main contributions are as follows:

1. We present a curated dataset of 8.5K grade school math questions and natural language solutions, useful for probing the informal reasoning ability of large language models.
2. We show that, compared to a finetuning baseline, the use of verifiers results in approximately the same performance boost as a 30x model size increase, and that verifiers scale significantly better with increased data.
3. We show that dropout acts as a strong regularizer, significantly improving performance for both finetuning and verification.

2 Dataset

GSM8K consists of 8.5K high quality grade school math problems created by human problem writers. We segmented these into 7.5K training problems and 1K test problems. These problems take between 2 and 8 steps to solve, and solutions primarily involve performing a sequence of elementary calculations using basic arithmetic operations ($+$ $-$ $\times$ $\div$) to reach the final answer. A bright middle school student should be able to solve every problem.

We created GSM8K based on the following design principles.

- **High Quality** We avoid error-prone scraping procedures and instead rely on human workers to create problems. After performing extensive quality control based on workers’ answer agreement, we estimate that less than 2 percent of problems contain breaking errors.
- **High Diversity** We strive for high diversity among problems. We actively avoid designing problems that are drawn from the same linguistic template or differ only in superficial details, an issue that is prevalent among many other datasets. By creating each individual problem to be relatively unique, held-out test performance becomes a far more relevant metric.
- **Moderate Difficulty** We choose a problem distribution that is challenging for large state-of-the-art language models, without being completely intractable. GSM8K will help us better understand the data scaling trends of different models and methods in this difficulty sweet spot. Problems require no concepts beyond the level of early Algebra, and the vast majority of problems can be solved without explicitly defining a variable.
- **Natural Language Solutions** We collect solutions in natural language rather than as pure math expressions. We believe this is the most generally useful data format, and we expect it to shed light on the properties of large language models’ internal monologues. We instructed problem writers to explain their work as much as possible, but we allowed them to write solutions in their own diverse linguistic styles.

The full GSM8K dataset can be found at <https://github.com/openai/grade-school-math>. Example problems are shown in Figure 1, and we discuss additional dataset details in Appendix A.

3 Related Work

3.1 Related Datasets

Early math word problem datasets (Kushman et al., 2014; Roy and Roth, 2015) are relatively small and are not well suited for testing the limits of modern language models. Dolphin18K (Huang et al., 2016) is a larger dataset containing